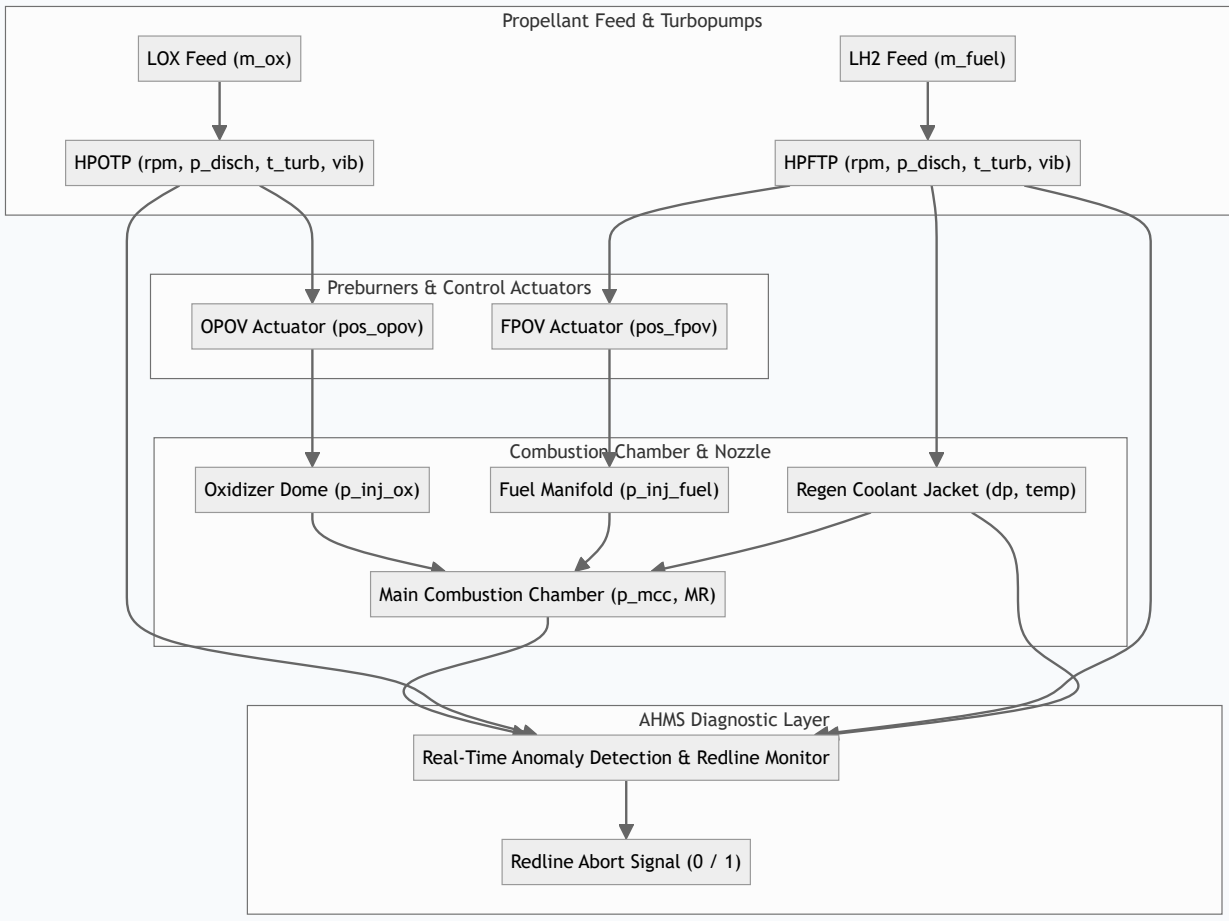
Liquid Rocket Engine Anomaly Detection Benchmark (LRE-AD-25K)

1. Dataset Overview

The **LRE-AD-25K** dataset contains high-frequency multivariate telemetry synthesized from empirical ground-test and flight data of staged-combustion liquid rocket engines (grounded in the **NASA Space Shuttle Main Engine / RS-25** architecture and contemporary Integrated System Health Management literature).

- **Location on Disk**: `rocket_engine_anomaly_dataset.csv`
- **Metadata Documentation**: `README.md`
- **Total Samples**: 25,025 time-series observations
- **Sampling Frequency**: 10 Hz ($\Delta t = 0.1$ s)
- **Test Firings**: 25 hot-fire runs (10 Nominal, 15 Anomaly runs covering 8 distinct failure modes)
- **Channels**: 26 synchronized features (21 physical sensor channels + 5 operational/diagnostic labels)



2. Sensor Channels & Telemetry Data Dictionary

Index	Field	Unit	Nominal (104% RPL)	Physical Description
0	run_id	-	-	Test firing run identifier (TEST_NOM_01 to TEST_ANO_INSTAB_01)
1	timestamp_sec	s	0.0 – 100.0	Elapsed test time from start of sequence
2	operational_phase	-	Mainstage_104	Operational regime (Ignition, Ramp_Up, Mainstage, Throttle, Shutdown)
3	throttle_pct	%	104.0%	Commanded power level relative to 100% Rated Power Level (RPL)
4	p_mcc_bar	bar	212.17 ± 0.62	Main Combustion Chamber static pressure (P_c)
5	m_ox_kg_s	kg/s	426.4 ± 0.8	Liquid Oxygen (LOX) total mass flow rate
6	m_fuel_kg_s	kg/s	70.7 ± 0.25	Liquid Hydrogen (LH2) total mass flow rate
7	of_mixture_ratio	-	6.03 ± 0.02	Oxidizer-to-fuel mixture ratio ($O/F = \dot{m}_{ox}/\dot{m}_{fuel}$)
8	p_hpotp_disch_bar	bar	318.4 ± 0.9	High-Pressure Oxidizer Turbopump discharge pressure
9	t_hpotp_turb_k	K	781.4 ± 1.8	HPOTP preburner turbine inlet temperature
10	rpm_hpotp	RPM	$28,660 \pm 45$	HPOTP rotational speed
11	vib_hpotp_radial_g	g_{rms}	3.6 ± 0.15	HPOTP bearing radial accelerometer vibration amplitude
12	p_hpftp_disch_bar	bar	454.2 ± 1.2	High-Pressure Fuel Turbopump discharge pressure
13	t_hpftp_turb_k	K	987.0 ± 2.2	HPFTP preburner turbine inlet temperature
14	rpm_hpftp	RPM	$35,795 \pm 55$	HPFTP rotational speed
15	vib_hpftp_radial_g	g_{rms}	4.6 ± 0.18	HPFTP bearing radial accelerometer vibration amplitude
16	p_inj_ox_dome_bar	bar	235.3 ± 0.7	Oxidizer injector dome static manifold pressure
17	p_inj_fuel_man_bar	bar	245.1 ± 0.8	Fuel injector manifold static pressure
18	coolant_dp_bar	bar	48.4 ± 0.3	Differential pressure (ΔP) across regenerative nozzle jacket
19	coolant_jacket_temp_k	K	546.2 ± 1.0	Exit coolant manifold temperature
20	pos_fpov_pct	%	55.1 ± 0.1	Fuel Preburner Oxidizer Valve position
21	pos_opov_pct	%	59.2 ± 0.1	Oxidizer Preburner Oxidizer Valve position
22	anomaly_flag	{0, 1}	0	Binary indicator (0 = Nominal, 1 = Fault active)
23	anomaly_class	String	NOMINAL	Fault taxonomy label (9 distinct classes)
24	severity	[0.0, 1.0]	0.0	Normalized anomaly severity
25	redline_tripped	{0, 1}	0	Redline safety boundary trip

3. Failure Mode Signatures & Physical Grounding

The anomaly taxonomy reflects documented failure modes from liquid rocket engine test stand and flight histories:

1. `HPOTP_BEARING_DEGRADATION`
   - *Physical Mechanism*: Ball/roller spall and bearing cage degradation under high radial loads in LOX environment.
   - *Telemetry Signature*: Sharp rise in HPOTP radial vibration amplitude ($g_{\text{rms}} > 8.5\ g$), localized friction heat, and subtle shaft rotational drag.
2. `HPFTP_TURBINE_BLADE_EROSION`
   - *Physical Mechanism*: Thermal cracking and aerodynamic profile loss on high-pressure hydrogen turbine blades.
   - *Telemetry Signature*: Drop in turbine efficiency; engine controller drives FPOV valve open to maintain turbopump power, spiking turbine inlet temperature above the $1080\ \text{K}$ redline.
3. `LOX_PUMP_CAVITATION`
   - *Physical Mechanism*: Pump inlet Net Positive Suction Head (NPSH) drops below vapor pressure, forming vapor cavities on impeller leading edges.
   - *Telemetry Signature*: Head breakdown causing a sudden 10–15% drop in discharge pressure, oxidizer mass flow sag, and hydraulic pressure ripple.
4. `COOLANT_CHANNEL_RESTRICTION`
   - *Physical Mechanism*: Particulate debris or wall deformation constricting milled copper cooling channels in the nozzle throat.
   - *Telemetry Signature*: Regenerative cooling jacket ΔP surges from 48 bar to $> 65\ \text{bar}$, exit coolant temperature surges past $640\ \text{K}$.
5. `FPOV_ACTUATOR_DRIFT`
   - *Physical Mechanism*: Electro-hydraulic servo-valve drift or mechanical feedback potentiometer bias.
   - *Telemetry Signature*: Actual valve position lags commanded profile by 10–14%, throwing preburner mixture ratio off balance and dragging down chamber pressure.
6. `INJECTOR_EROSION_OR_CHOKING`
   - *Physical Mechanism*: Coaxial injector element recess erosion or LOX post deformation.
   - *Telemetry Signature*: Manifold dome backpressure rises while combustion chamber pressure drops due to degraded characteristic exhaust velocity (c^*).
7. `SENSOR_BIAS_DRIFT_P_MCC`
   - *Physical Mechanism*: Transducer strain-gauge drift due to thermal gradient across the chamber sensor port.
   - *Telemetry Signature*: Chamber pressure reading drifts +22 bar higher, while independent turbopump speeds and flow rates remain unaffected (sensor-only fault).
8. `COMBUSTION_INSTABILITY_ROUGHNESS`
   - *Physical Mechanism*: Acoustic coupling between heat release and chamber 1st longitudinal (1L) acoustic resonance.
   - *Telemetry Signature*: Standard deviation and noise amplitude of P_c surges from $\pm 0.5\ \text{bar}$ to over $\pm 12\ \text{bar}$, accompanied by broad-spectrum turbomachinery vibration.

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4. Academic Publications & Literature References

> **NOTE**
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> All parameters, sensor relationships, redline trip criteria, and failure progressions in this dataset are modeled in direct alignment with the following peer-reviewed publications and NASA technical reports:

1. **Cikanek, H. A. (1986)**
   Characteristics of Space Shuttle Main Engine Failures.
   AIAA/ASME/SAE/ASEE 22nd Joint Propulsion Conference, Huntsville, AL. AIAA Paper 1986-1411.
   DOI: 10.2514/6.1986-1411
   Key Contribution: Classical taxonomy of SSME turbomachinery, preburner, and nozzle failure mechanisms.

2. **Schwabacher, M., Samuels, M., & Waterman, L. (2002)**
   NASA's Advanced Health Management System for the Space Shuttle Main Engine.
   Proceedings of the 2002 IEEE Aerospace Conference, Big Sky, MT, Vol. 6, pp. 2809–2818.
   DOI: 10.1109/AERO.2002.1035987
   Key Contribution: Definition of real-time vibration spectral thresholds, redline cutoffs, and HPOTP/HPFTP sensor anomaly detection logic.

3. **Duyar, A., & Merrill, W. C. (1992)**
   Fault Diagnosis for the Space Shuttle Main Engine.
   Journal of Guidance, Control, and Dynamics, 15(2), pp. 384–389.
   DOI: 10.2514/3.20850
   Key Contribution: Multi-variable state-space modeling of rocket engine component degradation (pumps, valves, injectors).

4. **Cha, J., Ha, C., & Ko, S. (2020)**
   Anomaly detection in liquid-propellant rocket engines using deep autoencoders with multivariate time series.
   Aerospace Science and Technology, Vol. 106, 106186.
   DOI: 10.1016/j.ast.2020.106186
   Key Contribution: Multivariate time-series autoencoder benchmarking on 75-ton-class liquid rocket engine hot-fire telemetry.

5. **Wang, J., Zhang, Y., Liu, H., & Cheng, Y. (2021)**
   A survey on liquid rocket engine fault diagnosis and health management.
   Acta Astronautica, Vol. 182, pp. 305–320.
   DOI: 10.1016/j.actaastro.2021.02.015
   Key Contribution: Comprehensive taxonomy of physical fault coupling, operational transients, and AI-driven diagnostic architectures.

6. **Cheng, Y., Wu, J., & Chen, Y. (2019)**
   Real-time fault detection of liquid rocket engine based on adaptive thresholding and deep neural network.
   IEEE Transactions on Aerospace and Electronic Systems, 55(4), pp. 1856–1868.
   DOI: 10.1109/TAES.2018.2878799
   Key Contribution: Methods for discriminating between sensor-only instrumentation faults (bias/drift) and genuine engine plant degradation.

7. **Bickford, R., & Davis, R. (2000)**
   Real-Time Rocket Engine Health Monitoring System.
   36th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, Las Vegas, NV. AIAA Paper 2000-3622.
   DOI: 10.2514/6.2000-3622
   Key Contribution: Valve actuator drift compensation, analytical redundancy, and hot-fire testing verification.

8. **Huzel, D. K., & Huang, D. H. (1992)**
Modern Engineering for Design of Liquid-Propellant Rocket Engines.
AIAA Progress in Astronautics and Aeronautics, Vol. 147. ISBN: 978-1-56347-013-4.
Key Contribution: Governing thermodynamic and turbomachinery fluid relationships ($P_c = \dot{m}c^*/A_t$, pump head rise, and cooling jacket heat flux).